5 **(a)** **(i)** Gravitational potential at a point in a gravitational field is the work done per unit mass by an external force in bringing a small test mass from infinity to that point.

(ii) Gravitational potential at infinity is zero.

Since gravitational force is attractive in nature, to bring a mass from infinity to a point in the gravitational field, the direction of the external force is opposite to the direction of displacement of the mass. This results in negative work done per unit mass by the external force.

Hence, based on its definition, gravitational potential is a negative value.

(b) **(i)**

$$\Delta E_P = \left(-\frac{GM_E m}{x}\right) - \left(-\frac{GM_E m}{R_E}\right)$$

$$= GM_E m\left(\frac{1}{R_E} - \frac{1}{x}\right)$$

$$= \left(6.67\times10^{-11}\right)\left(6.0\times10^{24}\right)(1600)\left(\frac{1}{6400\times10^3} - \frac{1}{\left(6400\times10^3\right)+\left(2.1\times10^7\right)}\right)$$

$$= 7.6681\times10^{10} = 7.67\times10^{10}\ \text{J}$$

(ii) **1.** Gravitational force provides the centripetal force on the satellite.

$$\frac{GM_E m}{r^2} = \frac{mv^2}{r} \quad \text{where } m \text{ is the mass of the satellite}$$

$$v = \sqrt{\frac{GM_E}{r}}$$

2. By the principle of conservation of energy, if the satellite has just enough energy to escape to infinity, its total energy is zero.

Let E_{K1} be the kinetic energy of satellite just after the boost.

$$E_P + E_{K1} = 0$$

$$-\frac{GM_E m}{r} + E_{K1} = 0$$

$$E_{K1} = \frac{GM_E m}{r}$$

Just before the boost, kinetic energy of satellite in orbit,

$$E_K = \frac{1}{2}mv^2$$

$$= \frac{1}{2}m\left(\sqrt{\frac{GM_E}{r}}\right)^2$$

$$= \frac{GM_E m}{2r}$$

$$\text{ratio} = \frac{E_{K1}}{E_K} = \frac{GM_E m}{r} \bigg/ \frac{GM_E m}{2r} = 2$$

6 (c) (i) Effective resistance of Q and LDR